

CARLOS ANDRÉS MONTERO MORENO

Backend Developer | Node.js / TypeScript | Linux | REST APIs | CI/CD

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Portfolio: zamoxv.github.io/cv-carlos-montero | GitHub: github.com/zamoxv | LinkedIn: linkedin.com/in/carlosmontero-dev

PROFESSIONAL SUMMARY

Backend developer focused on Node.js and TypeScript, with experience designing and building REST APIs, authentication systems and applications deployed on Linux, applying layered architecture, error handling and development best practices. Ongoing training in AI-assisted development and agents. My technical profile is complemented by 10+ years in the energy industry operating critical systems, which brings analytical thinking, discipline, problem-solving under pressure and self-directed learning — skills I apply today to managing my own infrastructure and to freelance development projects.

TECHNICAL SKILLS

Development

- Backend — Node.js, Express.js, REST APIs
- Languages — JavaScript (ES6+), TypeScript, Python (basic), Bash
- Databases — PostgreSQL, MongoDB, CouchDB
- Dev Tools — Git, GitHub, GitHub Actions, CI/CD, GitHub Pages
- Concepts — Layered architecture, authentication, API integration, automation
- Fundamentals of AI-assisted development and agents

Systems & Infrastructure

- Linux (Fedora, Ubuntu Server) — installation, configuration and maintenance
- Self-hosted server administration: Samba, Jellyfin, AdGuard Home, backups
- Deployment and installation automation with Bash
- Hardware/software troubleshooting and diagnostics
- Analytical thinking and problem-solving
- Clear communication and collaboration in multidisciplinary teams

PROJECTS

- **Landing page and CRM for P&Z Energía e Innovación** (Synware, freelance) — Node.js, Express and TypeScript, for a residential solar panel company.
- **Jasjic – Shared Expenses App**: web app with a Node.js/Express backend, Google OAuth authentication and CouchDB persistence, deployed in production on Linux (Raspberry Pi).
- **Homelab Installer**: TUI installer to set up and maintain a home server on Ubuntu Server, with a plugin system, power profiles, backup/migration (including remote migration over SSH) and system health diagnostics.
- **REST API – Task Manager**: backend with Node.js, Express and TypeScript, layered architecture and PostgreSQL persistence.
- **CV Builder – Automation**: PDF CV generation with Pandoc, LaTeX and Bash, integrated with GitHub Actions (CI/CD).
- **API Explorer (SPA)**: web application consuming public APIs, with dynamic data rendering and state management.

PROFESSIONAL EXPERIENCE

Backend Developer (Freelance) — Synware

Remote | 2025 – Present

- Develop web applications and REST APIs with Node.js, Express and TypeScript, implementing authentication and deploying on Linux environments.
- Designed and developed a landing page and CRM for clients, including P&Z Energía e Innovación.

Field Operator — Catalytic Cracking Division — ENAP Aconcagua Refinery (Chile's national oil company)

Concón, Chile | Jan 2016 – Aug 2025

- Operated and maintained critical industrial units (LPG treatment, Merox extraction, amine regeneration, sulfur recovery, among others) under strict safety, environmental and quality standards.
- Monitored and controlled process variables, diagnosed failures, wrote technical reports and coordinated with multidisciplinary teams in a demanding 24/7 operating environment.

IT Support Technician — Self-employed

Chile | 2006 – Present

- Hardware and software troubleshooting, operating system installation and end-user technical support.

Process and Quality Control Technician — CCTVal | Universidad Técnica Federico Santa María

Valparaíso, Chile | 2013 – 2014

- Took part in a process and quality control research project; performed advanced polishing for the Light Guides project for a particle accelerator at Jefferson Lab (Newport News, Virginia, USA).

EDUCATION

- B.S. in Computer and Information Engineering — Universidad Andrés Bello, 2022
- Electrical and Electronics Technician (Associate degree) — DUOC UC, 2014

CERTIFICATIONS & CONTINUING EDUCATION

- AI Development Initiation Certificate (AI Development: Programming with Agents) — BIG School / MoureDev, Jul 2026

LANGUAGES

- Spanish — Native
- English — TOEIC certified, professional working proficiency